

10.3K

Total Shipments



1.2K

Total Late Deliveries



240.1

Average Insurance C...



21.9

Average Pack Price (i...



# End-to-End Supply Chain Insights & Optimization

## Operational Performance, Risk Compliance, and Procurement Audit

# Logistics Audit

10.3K

Total Shipments

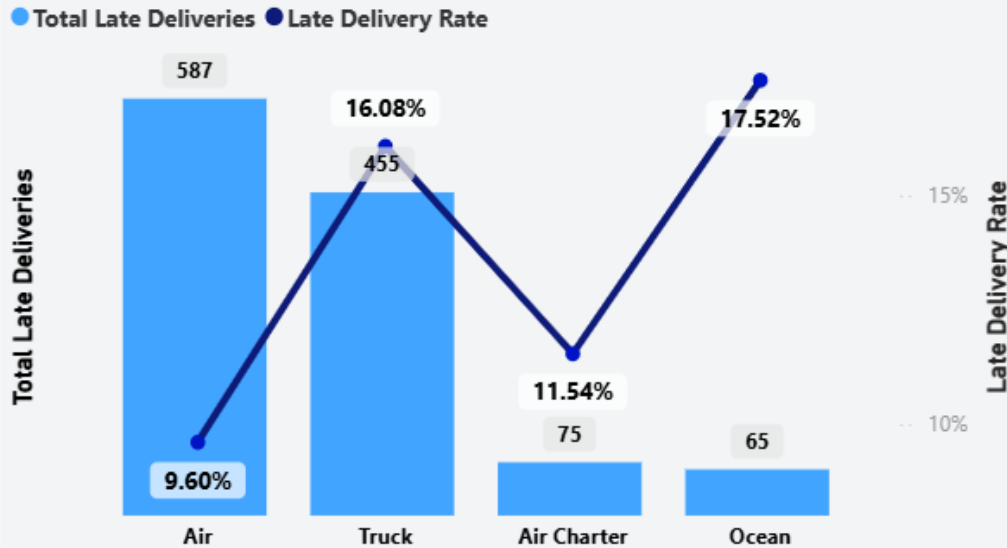


1.2K

Total Late Deliveries



- **Total Volume:**  
10,300 shipments successfully audited across the global network.
- **Service Gap:**  
Identified a baseline late delivery rate of 11.6% (1.2K units).
- **Observation:**  
While Air has the highest raw count of late deliveries (587), Ocean has the highest late rate (17.52%).
- **Recommendation:**  
If the delivery is time-sensitive, the data suggests Air is the most reliable (9.60%), while Ocean remains the least reliable mode with a 17.52% late rate.

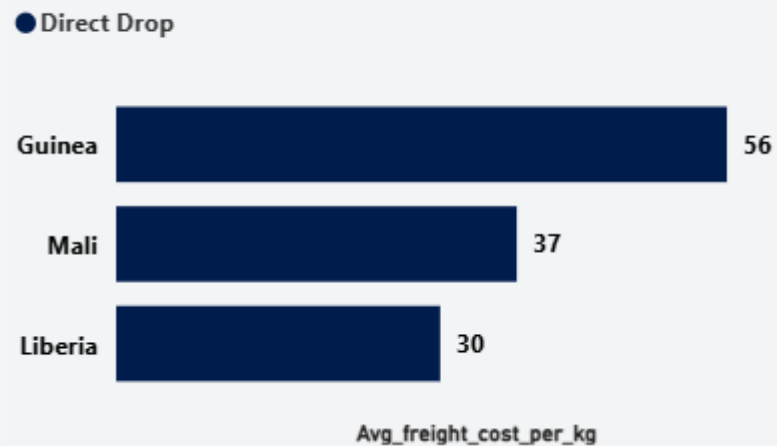


# Regional Freight Cost Analysis

Global Freight Cost Distribution



Showing Top 3 countries (All shipped via Air & Direct Drop)



- **Observation:**  
Freight costs are heavily skewed towards West African destinations when utilizing Air and Direct Drop modes.
- **Specific Insight:**  
Guinea maintains the highest average freight cost per kg (56), significantly out-spacing other regional hubs, followed by Mali (37) and Liberia (30).
- **Recommendations:**
  1. For time-insensitive goods, negotiate a transition to Ocean Freight or Regional Trucking.
  2. Conduct a vendor-specific audit in these three countries to identify lower-cost shipment alternatives.

## Vendor Efficiency & Cost Correlation

| Country | Vendor                     | Avg_freight_cost_per_kg (in usd) |
|---------|----------------------------|----------------------------------|
| Liberia | Trinity Biotech, Plc       | 42.24                            |
| Liberia | Standard Diagnostics, Inc. | 25.77                            |
| Guinea  | Orgenics, Ltd              | 56.38                            |
| Liberia | Orgenics, Ltd              | 29.82                            |
| Mali    | Orgenics, Ltd              | 33.97                            |
| Mali    | Orasure Technologies Inc.  | 132.01                           |



- Observations:**

1. There is a massive \$96/kg price gap between Orgenics, Ltd & Orasure Technologies Inc. (nearly 4 times) in Mali.
2. Standard Diagnostics, Inc. is currently the most cost-effective partner for Liberia.
3. There is an inverse relationship between the weights and freight cost: “As the weight in kg increases, average freight cost per kg almost always goes down”.

- Recommendations:**

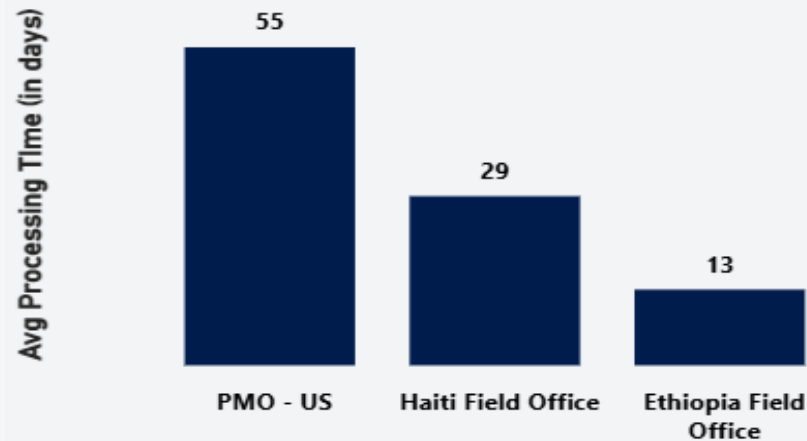
1. Audit Orasure Inc. contracts to identify cold chain requirements vs. vendor overcharging.
2. Implement batch shipping for low-weight cargo to capture 66%+ cost savings through consolidation.

# Operational Velocity & Bottleneck Analysis

Processing Bottlenecks by Mode

| Management Hub        | Shipment Mode | Avg Processing Time (in days) |
|-----------------------|---------------|-------------------------------|
| Ethiopia Field Office | Truck         | 13                            |
| Haiti Field Office    | Air           | 29                            |
| PMO - US              | Air           | 30                            |
| PMO - US              | Ocean         | 33                            |
| PMO - US              | Air Charter   | 44                            |
| PMO - US              | Truck         | 159                           |

Operational Velocity by Management Hub



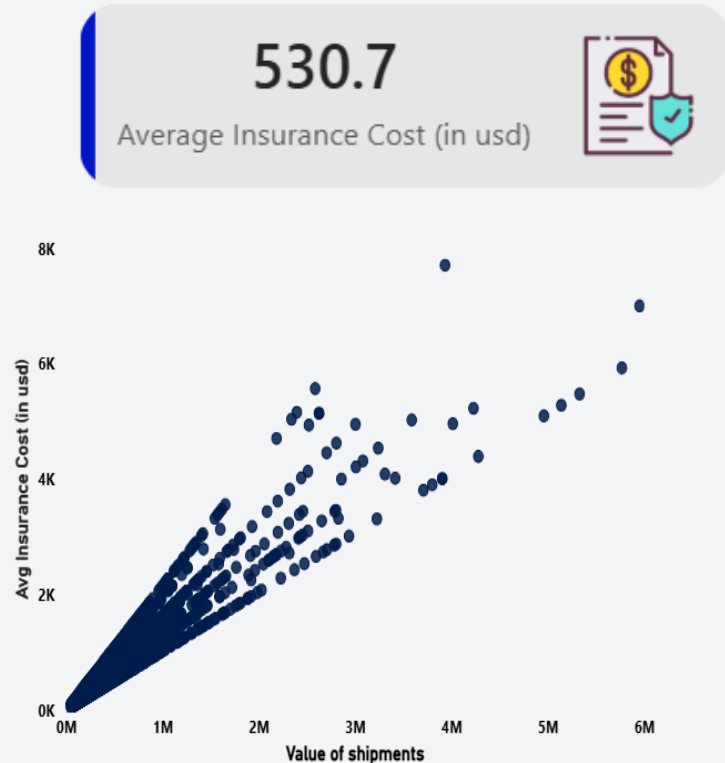
- Observations:**

- 1. US Trucking Bottleneck:** The PMO-US Management Hub is experiencing a severe bottleneck in its trucking operations, with a 159-day average processing time.
- 2. Comparative Efficiency:** The Ethiopia Field Office is the most agile hub, processing shipments in just 13 days – over 12 times faster than the US trucking segment.

- Recommendations:**

- 1. Immediate Operational Review:** Prioritize PMO-US lane audit to isolate customs, capacity, or carrier bottlenecks.
- 2. Knowledge Transfer:** Scale Ethiopia's agile workflow to Haiti and US offices.

# Risk Compliance & Insurance Value



- **Observations:**

1. The average insurance cost for shipments valued over \$50000 is \$530.7.
2. There are 113 high-value shipments (up to \$1.08M) currently in the system with no recorded insurance coverage.
3. Analysis confirms a linear relationship between cargo value and insurance premiums, indicating a standardized risk mitigation strategy.

- **Recommendations:**

1. Flag 113 uninsured IDs for Finance to verify third-party coverage or active exposure.
2. Mandate 'Hard Stop' in procurement for POs > \$50K with unverified insurance.

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Average Insurance Cost (in usd)



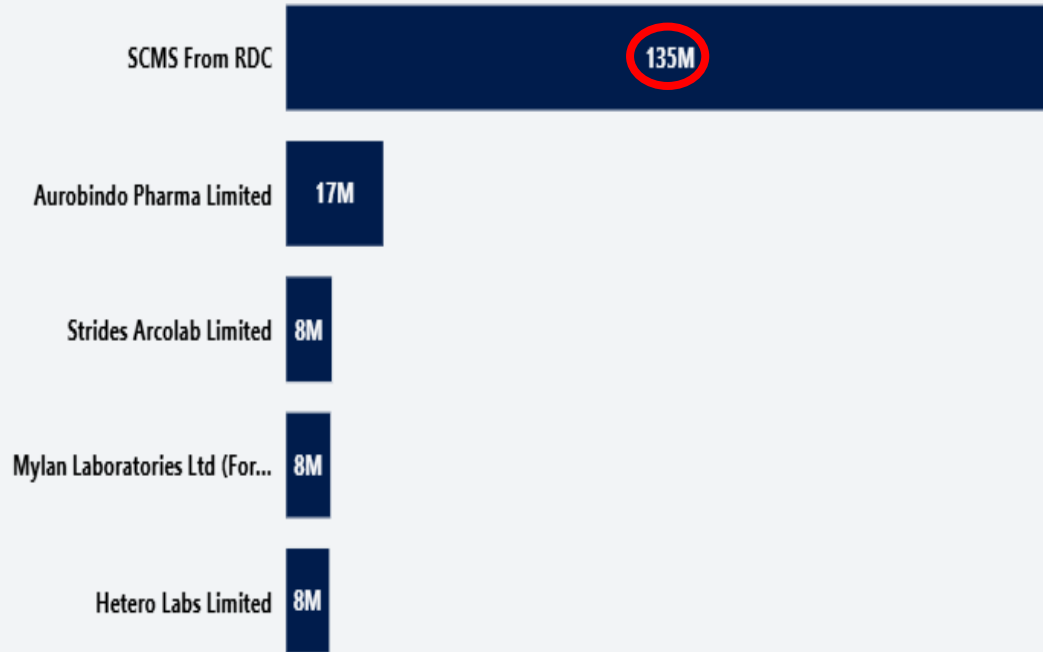
113.0

Total Shipments



# Vendor Performance

Total Quantity Handled by Vendor



Total Quantity Handled

- **Observations:**
  1. **Market Dominance:** SCMS From RDC is the primary strategic partner, handling 135M units – which is more than 75% of the total volume handled by top 5 vendors combined.
  2. **Tier 2 Partners:** Aurobindo Pharma Limited (17M) and Strides Arcolab Limited (8M) serve as critical secondary supports, though their volume is significantly lower.
- **Recommendations:**
  1. **Risk Mitigation:** Establish contingency protocols for SCMS From RDC to ensure continuity during vendor strikes or failures.
  2. **Volume Leveraging:** Leverage 135M unit volume to negotiate superior freight rates and priority processing.

# Integrated Supply Chain Control Center



“An end-to-end visualization suite enabling real-time monitoring of global logistics, vendor performance, and risk compliance.”

# Project Impact & Strategic Summary

## **Operational Efficiency**

Identified a 159-day lag in US Trucking; proposed a workflow audit based on the 13-day Ethiopia model.

## **Cost Optimization**

Highlighted West Africa as a high-cost hub and flagged a \$96/kg price gap in Mali.

## **Risk & Compliance**

Uncovered 113 high-value shipments without insurance; proposed a “Hard Stop” procurement protocol.

*“This audit establishes a data-driven framework for future operational improvements and risk mitigation strategies.”*

# Thank You For Your Time!

## Sanskriti Agrawal

“Data Analyst | Power BI & Risk Mitigation Specialist”



**Connect for Collaboration**

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**Interact with the Live Dashboard**

[https://www.novypro.com/profile\\_about/1778215499097x558459591747214600](https://www.novypro.com/profile_about/1778215499097x558459591747214600)



**View SQL Queries & Documentation**

<https://github.com/SanskritiAgrawal123>

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